CS & IT ENGINEERING

THEORY OF COMPUTATION

Pushdown Automata

Lecture – 01

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Recap of Previous Lecture

Topic ?????

Grammar Construction

types of Grammar

Ambiguous Grammar

Simplification of Grammar.

Topics to be Covered

Topic Push down automat

Topic ??

Topic ??

Topic ??

$L = \{a^n b^n \mid n \geq 1\}$ ⇒ Pushdown Automata

Dependency

DFA ×

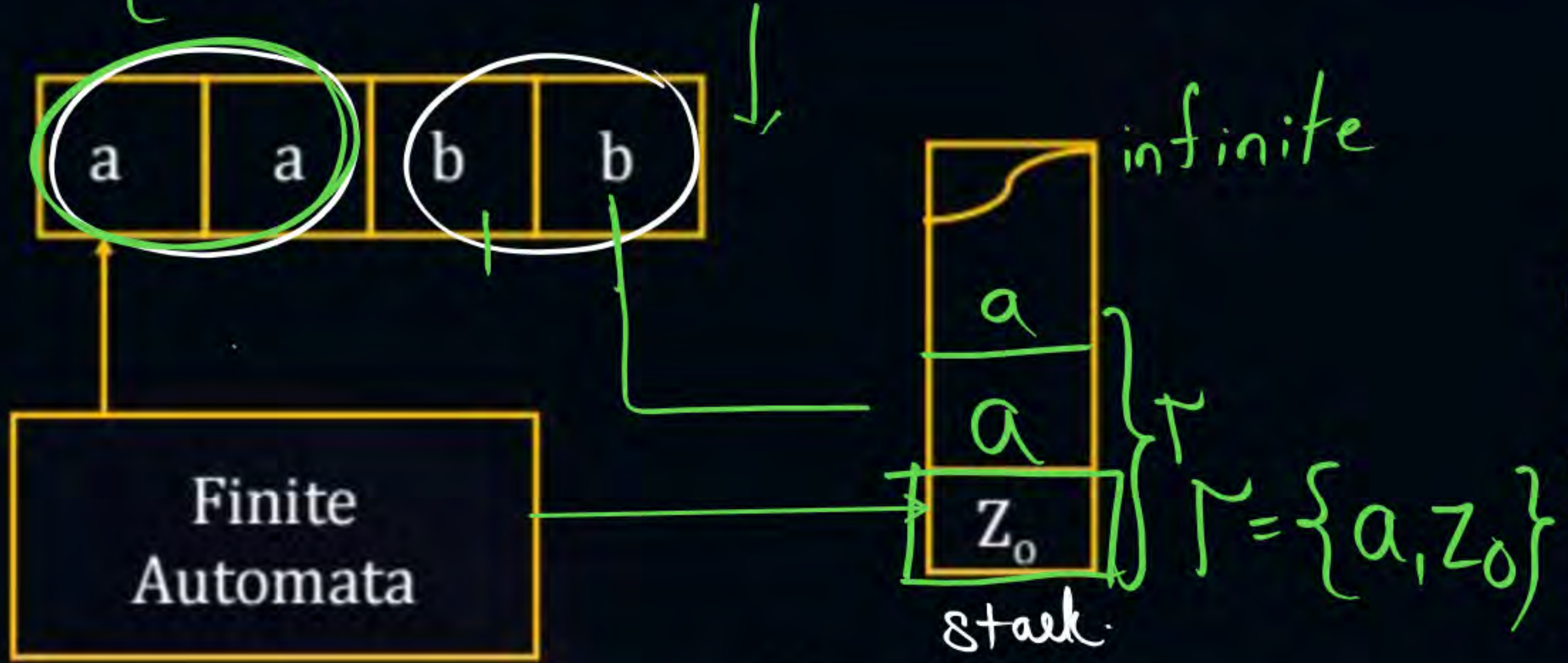
NFA ×

ε-NFA ×

Pushdown Automata (PDA)

$a^{10} b^{10}$

$\{a^n b^n\}$



① Push

② Pop

F.A. + Stack = PDA

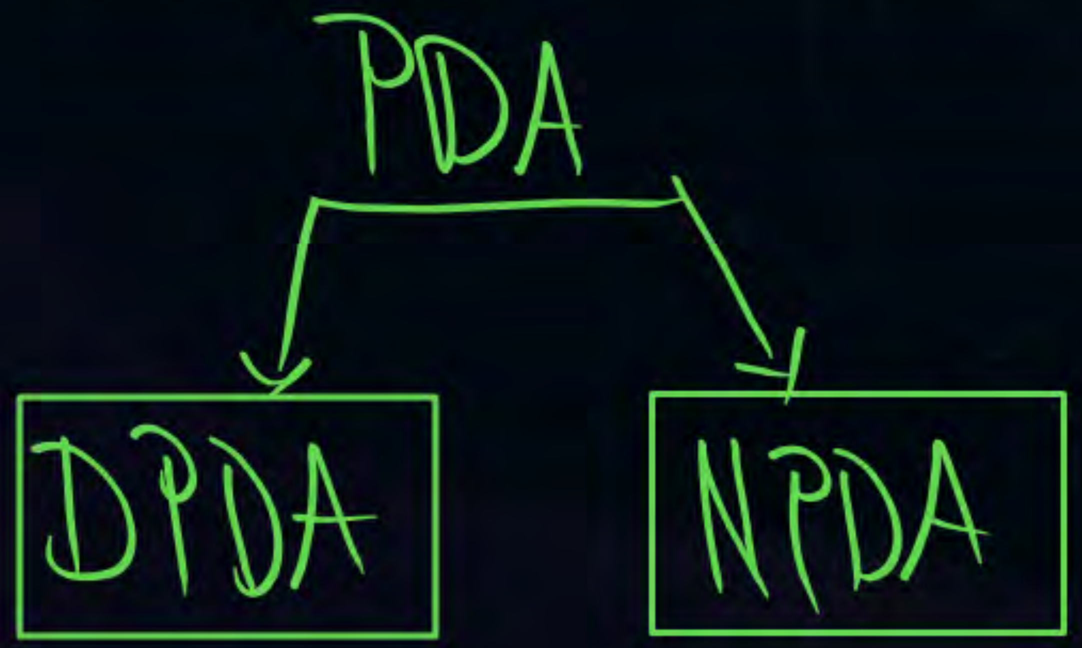
$L = \{a^n b^n c^n \mid n \geq 1\}$

PDA not possible

F.A < PDA
(more)

Topic : PDA

- 1 Finite Automata having additional power form of stack known as Push down automata.
- 2 Size of stack in Push Down automata is infinite
- 3 There exist only one type of push down automata i.e. “language recognisor”
- Push down automata can accept language in deterministic way or non-deterministic way



PDA $(Q, \Sigma, \delta, q_0, F, Z_0, \Gamma)$

Q :- Finite number of states

Σ :- Input alphabet

q_0 :- initial state

F :-set of final states

Z_0 :-initial stack symbol

Γ :-stack alphabet

δ :- transition function

$$\underbrace{Q}_{\text{state}} \times \underbrace{\Sigma \cup \{\varepsilon\}}_{\text{i/p}} \times \underbrace{\Gamma}_{\text{stack}} \rightarrow Q \times \Gamma^*)$$

$$\delta(q_1, a, Z_0) = (q_2, aZ_0)$$

$$\delta(q_1, a, z_0) = (q_1, \epsilon)$$

Formal Definition

F.A

PDA $(\boxed{Q, \Sigma, \delta, q_0, F}, \boxed{Z_0, \Gamma})$

Q :- Finite number of states

Σ :- Input alphabet

q_0 :- Initial State → only one

F :- Set of final states

Z_0 :- Initial stack elements

Γ :- Stack alphabet

δ :- transition function

$$\boxed{Q \times \Sigma \cup \{\varepsilon\} \times \Gamma \rightarrow Q \times \Gamma^{*})}$$

Topic : Note:

Note:- The following operation possible with PDA stack.

Push operation:- Moving i/p symbol from i/p buffer stack.

POP operation:- removing element from stack.

By pass operation:- $\boxed{\text{don't push \& don't pop}}$ (just reading symbol only)

SKIP

Topic : Empty Stack

By reading the string from left to Right by end of the string, if stack of the PDA is empty, then given string is accepted and irrelevent of No of final state.

Topic : Final State

By reading the string from left to right, end the string PDA enters into final state then given string is accepted and irrelevent about stack is empty or not.

Topic : Note

Note:- Number of language accepted by empty stack method and final state method is same in PDA.

The language L is accepted by empty stack if and only if L should be final state.

Topic PDA

PDA
less DPDA < NPDA (more)

F.A
DFA NFA

- 1 The expressive power of NPDA is more than DPDA.
- 2 By Default PDA means NPDA.
- 3 PDA practically used in compilers as parser.
- There are two types of acceptance method in PDA they are acceptance by empty stack and acceptance by final stack.

Notations:-

- Transition diagram
- Transitions

PDA (Acceptor)

- DPDA
- NPDA

Topic : Pushdown Automata

Construct Pushdown Automata for

$Q \times \boxed{\Sigma \cup \{\epsilon\}} \times \Gamma \rightarrow Q \times \Gamma^*$

① $\delta(q_0, a, Z_0) = (q_0, aZ_0)$ } Push a's

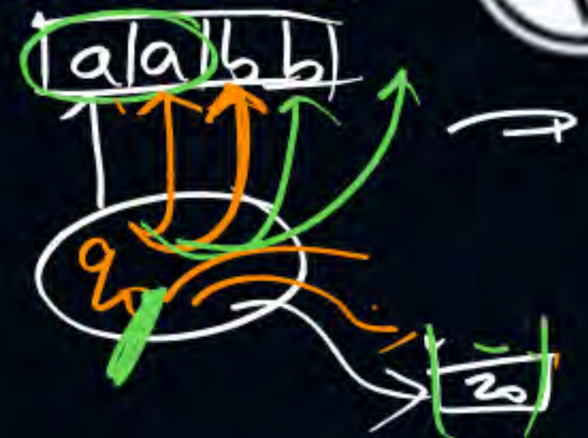
② $\delta(q_0, a, a) = (q_0, aa)$

③ $\delta(q_0, b, a) = (q_1, \epsilon)$ } Pop a's for b's

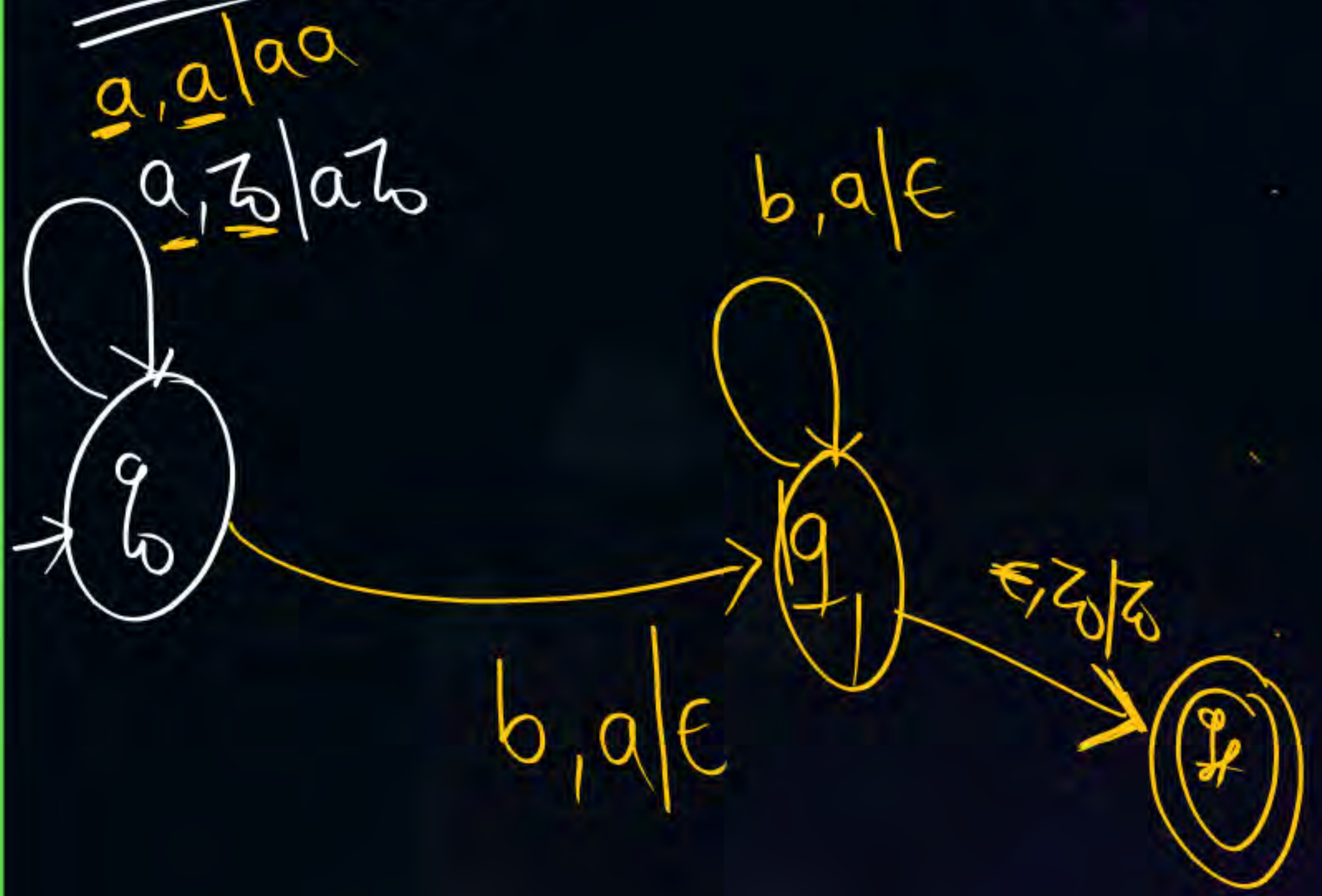
④ $\delta(q_1, b, a) = (q_1, \epsilon)$

⑤ $\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$ } accept

$L = \{a^n b^n \mid n \geq 1\}$

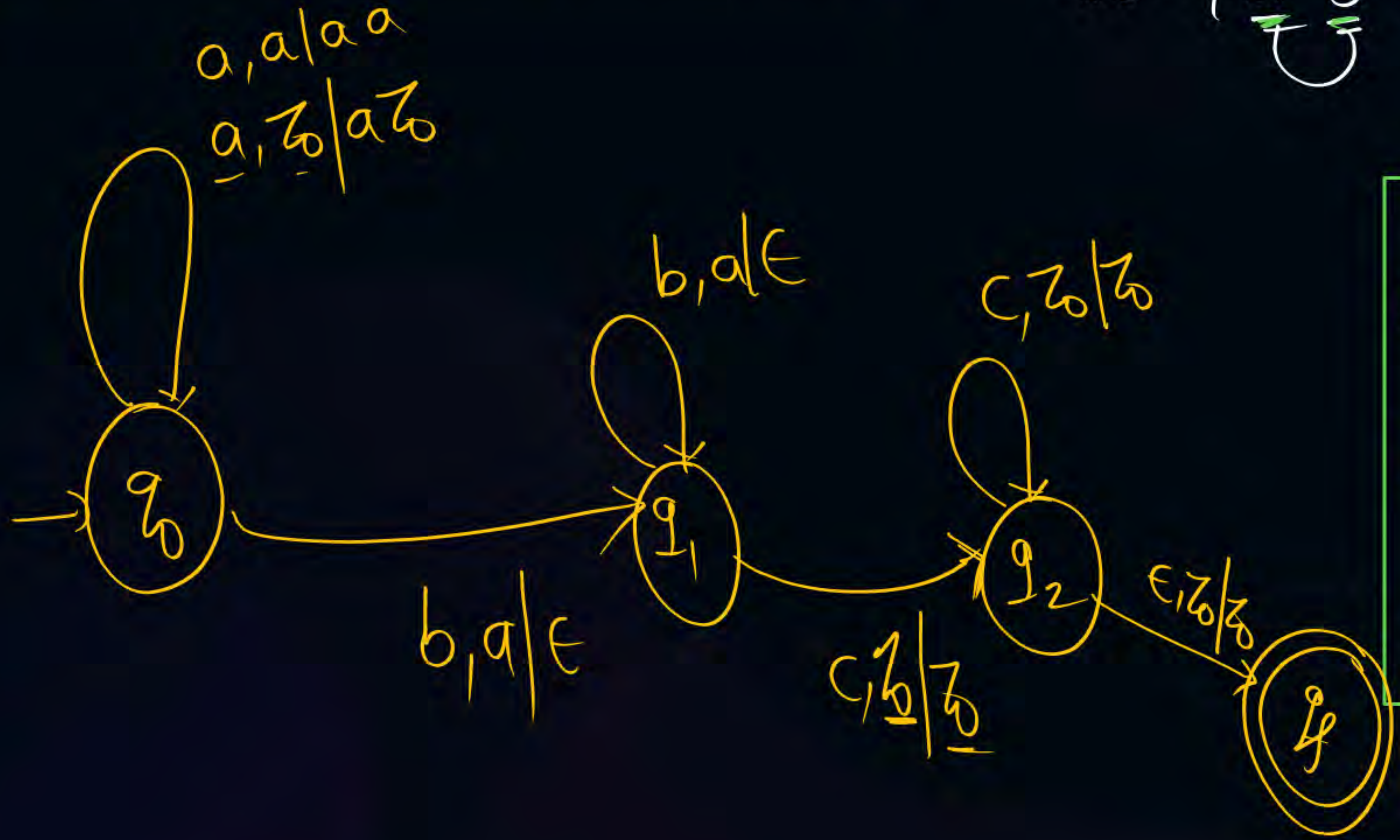


Transition diagram



Construct Pushdown Automata for

$L = \{a^n b^n c^m \mid n, m \geq 1\}$



Logic

1. a's ⇒ Push
2. b's ⇒ pop a
3. c's ⇒ SKIP

Topic : CONTEXTFREE LANGUAGE

Home Work

$L_1 = \{a^{n+m} b^n c^m \mid n, m \geq 1\}$

$L_2 = \{a^n b^{n+m} b^m \mid n, m \geq 1\}$

THANK - YOU